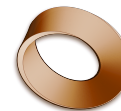


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



Refraction Investigation

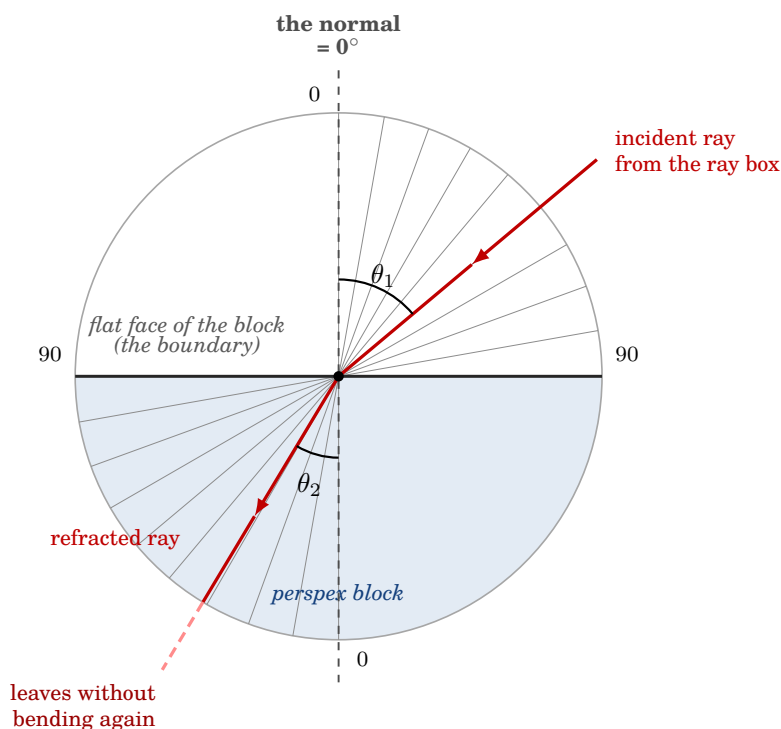
Criterion B — Inquiring and designing • Criterion C — Processing and evaluating

Learning intentions

You have already studied **reflection** — what light does when it *bounces off* a surface. This investigation is about **refraction** — what light does when it *passes into* a new material. Between them, these two are the laws that explain almost everything light does: every mirror, every lens, every pair of glasses, every camera.

Light changes direction when it crosses from one material into another. That part is easy to see. The harder question, and the one you are going to answer, is **how much** it bends — and what decides that.

Your apparatus: a ray box, a semicircular perspex block, and a printed protractor card. The block sits in the blue half of the card with its flat face along the centre line. Aim the ray at the **centre** of the card, read the **top** scale for the angle going in and the **bottom** scale for the angle coming out — and remember that on this card **the normal is 0°**.



Safety

The ray box gets **hot**. Keep paper, sleeves and pencil cases away from it, and do not touch the bulb. If you smell burning, switch it off at the power supply and tell Mr. Rawson. You will write this in your own words in the **Safety** part of B3.

B1 Your research question

You already know that light bends when it goes into a new material. So that is not the question. The question is **how much** it bends — and what you can do to change that.

Start by listing everything you can think of that might change **how much** the light bends. Write down every idea — there are no wrong ones at this stage.

*Last year's list: whether the edge of the block is flat or curved; how big the block is; how bright the light is; **the angle of incidence**; the temperature of the block or the air; what the block is made of; the colour of the light.*

One of those ideas is the easiest of all to change, because the protractor card is built for it: you can simply **swing the ray box round** and the light arrives at a different angle. Pick that one, and turn it into your research question.

Write it like this

How does changing _____ affect _____ ?
what you change *what you measure*

The first blank is your **independent variable** — the thing you change. The second is your **dependent variable** — the thing you measure.

Write your research question here.

How does changing the angle of incidence affect the angle of refraction?

B2 Your hypothesis

Your hypothesis is your **answer** to the research question, written before you do the experiment.

Write it like this

As the _____ increases, the _____
will **increase** / **decrease** / **stay the same**,
because _____ .

The word ‘because’ is where the marks are. A prediction on its own scores in the bottom band even when it turns out to be right. The reason is what lifts it.

Write your hypothesis here.

As the angle of incidence increases, the angle of refraction will also increase, because a bigger angle at the boundary means a bigger difference between the two sides of the ray, so the ray swings round further. The angle of refraction will always stay smaller than the angle of incidence, because light slows down going into perspex and so bends towards the normal. [Last year’s class predicted the refraction angle would decrease, and their data showed them otherwise — which is a perfectly good hypothesis. A hypothesis is not a guess at the right answer; it is a reasoned prediction you then test.]

B3 Your method

Draw your set-up

Draw the apparatus **as it will actually look** on the bench, and label every piece of equipment.

*A drawing showing the **ray box** (with its single-slit slide) and **power supply**, the **protractor card**, and the **semicircular perspex block** with its flat face along the centre line — all four labelled, and the ray arriving at the centre of the card.*

The independent variable — what you change

(i) Which values will you use?

For example

“I will use angles of incidence of 0° , 10° , 20° , 30° , 40° , 50° , 60° , 70° and 80° .”

I will use angles of incidence of 0° to 80° , going up in steps of 10° — nine values in total.

(ii) What will you measure it with? Name the instrument and its resolution.

Resolution — the word that earns the mark

Resolution is the **smallest change an instrument can show** — on a metre ruler marked in millimetres, 1 mm.

“I will use a _____ with a resolution of _____.”

I will use the protractor printed on the card, which has a resolution of 1° .

The dependent variable — what you measure

The same two things again: what you measure, and what with — including its resolution.

I will measure the angle of refraction using the lower protractor printed on the same card, which also has a resolution of 1° .

Control variables — what you keep the same

A **control variable** is something you keep the same so the test is fair. For each one write **three** things: **what it is**, **how you will keep it the same**, and **why it matters**.

A full example, all three parts

“I will keep **the temperature** the same. The classroom temperature changes very little, so doing the whole experiment in one lesson controls it. If the temperature changed, the perspex might refract light slightly differently.”

Choose only the ones that would actually change your result. Two or three real ones beat a long list.

***The same block throughout** — I will use one block for every reading. A different block might be a different material or shape, and would refract by a different amount. **The same ray box and the same power setting** — this keeps the brightness and the colour of the light the same, and different colours refract by slightly different amounts. **The ray hitting the centre of the card every time** — if the ray misses the centre it is no longer hitting the flat face along a radius, and both angles would be read from the wrong point. **One lesson** — this keeps the room temperature essentially constant.*

Repeats

Write it like this

"I will repeat each measurement **3 times** and find an **average**."

I will repeat each measurement 3 times and find an average, so that one misread value does not change my result.

Safety

Name something that could **genuinely cause injury**, and say what you will do about it.

The shape of a safety point

"_____ could cause _____. To stay safe I will _____."

The ray box gets hot and could cause burns, and paper left on it could catch fire. I will keep paper, sleeves and pencil cases away from the ray box, and I will not touch the bulb. If anything smells of burning I will switch it off at the power supply and tell the teacher.

Read it back

The two checks that decide the top band:

- ☐ Another Year 8 could follow my method and get the same data.
- ☐ My method would actually work.

C1 Your results

Three conventions, and they carry marks on their own:

- **Units go in the heading**, not beside each number. Away from a table, the unit goes next to the number.
- The **independent variable goes in the first column**.
- Work out each **average** and write it in.

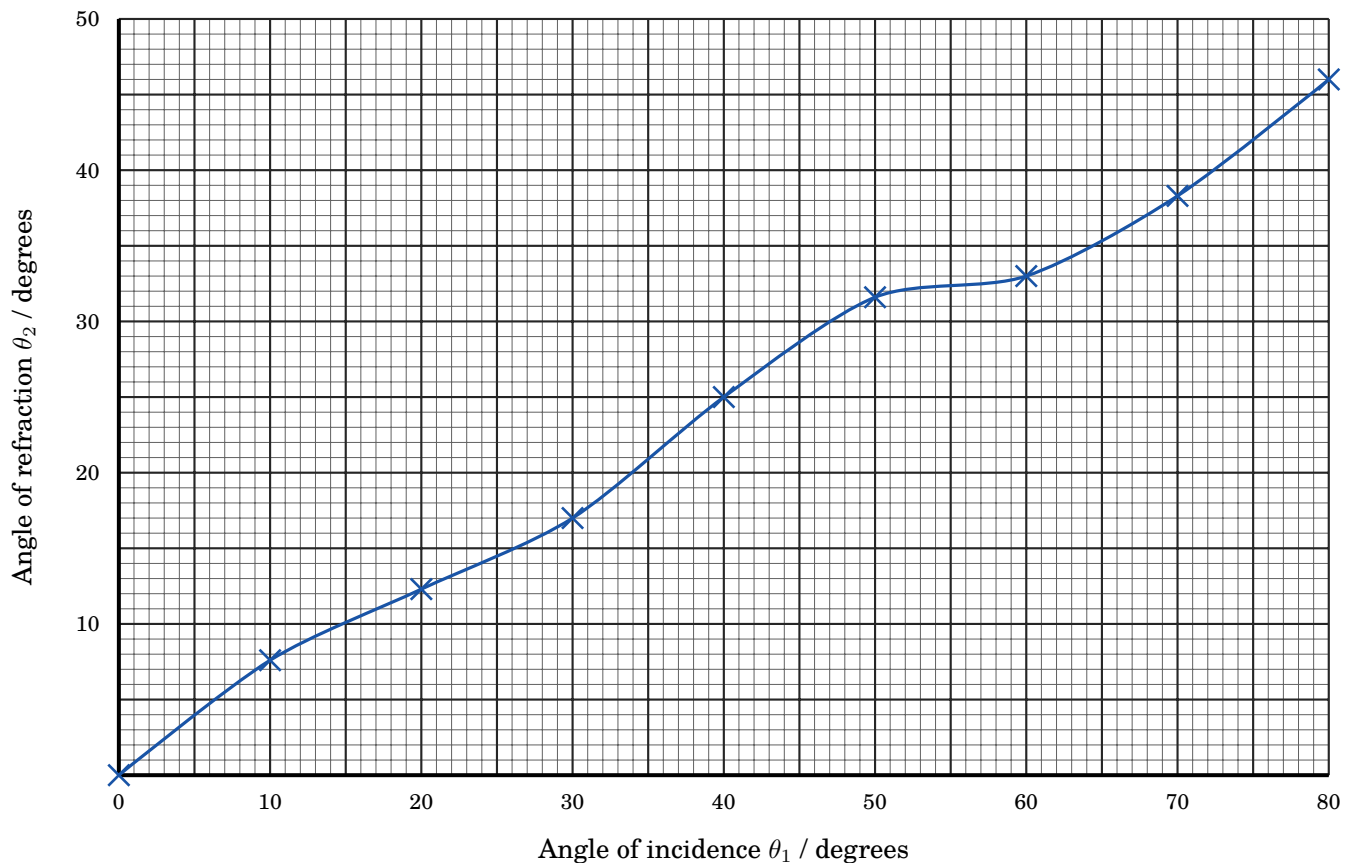
Angle of incidence θ_1 / degrees	Angle of refraction θ_2 / degrees			
	1st trial	2nd trial	3rd trial	Average
0	0	0	0	0
10	7	8	8	7.6
20	12	13	12	12.3
30	17	17	17	17
40	25	26	24	25
50	32	31	32	31.6
60	33	34	32	33
70	38	39	38	38.3
80	46	45	47	46

These are **last year's real averages** from the source document, not textbook values — which is why 60° and 80° sit off the smooth curve. Perspex ($n = 1.49$) predicts about 7, 13, 20, 26, 31, 35, 39, 41. A class getting within two or three degrees has measured well. **The 0° row reading 0° is the check worth making at reading one** — a table that starts at 90° has been read off the wrong scale.

Your graph

Plot the **average** angle of refraction (up) against the angle of incidence (across). Use a sharp pencil, then join your points with **one smooth curve**.

Both variables are numbers, so this is a **scatter graph with a line of best fit**. (You would only use a bar chart if the thing you changed were words rather than numbers.)



Four things the marker looks for:

- ☐ Axis numbers evenly spaced.
- ☐ Both axes labelled, with units.
- ☐ Points plotted accurately.
- ☐ A ruler used for the axes.

C2 Your conclusion

Your conclusion **answers your research question**. Look at the shape of your graph and choose the sentence that matches it.

Choose the sentence that fits your graph

Rising	As x increases, y increases.
Straight, through (0, 0)	x is directly proportional to y , because the graph is a straight line through the origin.
Straight, missing (0, 0)	There is a linear relationship between x and y , because the graph is a straight line that misses the origin.
Curving upwards	As x increases, y increases at an increasing rate.
Levelling off	However, after x is 20, y stops increasing.
No pattern	There is no correlation between x and y .

Then say **why you can say that**, in one of these three ways:

- **Numbers off your own graph** — “because when θ_1 was 20° , θ_2 was 13° , and when θ_1 was 40° , θ_2 was 26° .”
- “because the graph has a positive gradient.”
- “because the graph is curving upwards.”

Write your conclusion here.

As the angle of incidence increases, the angle of refraction also increases. When the angle of incidence was 20° the angle of refraction was about 12° , and when the angle of incidence was 40° the angle of refraction was about 25° . The angle of refraction was smaller than the angle of incidence every time.

To go further — explain it

Say **why** the physics does this. The best conclusions do not only describe the graph; they explain what is happening to the light.

Light travels more slowly in perspex than in air. The side of the ray that reaches the boundary first slows down first while the other side is still going quickly, so the ray swings round towards the normal. The bigger the angle of incidence, the bigger the head start between the two sides, so the more the ray turns.

C3 Was your hypothesis right?

This asks whether the prediction you made in **B2** turned out to be correct.

Write it like this

My hypothesis was **correct** / **partly correct** / **not correct**,
because _____.

Then point back at your conclusion and your data. **A hypothesis that turned out to be wrong is not a problem** — saying so clearly, and showing which data proves it, is what earns the marks here.

My hypothesis was correct. I predicted that as the angle of incidence increased the angle of refraction would also increase, and my results show exactly that: at 20° the refraction angle was 12° and at 60° it was 33° .

C4 Was your method valid?

A method is **valid** if it gave you results you can trust. Work down the left-hand column and tick everything that applies to your experiment.

My method <i>was</i> valid if...	My method <i>was not</i> valid if...
☐ The points are close to the line of best fit	☐ The points are a long way from the line of best fit
☐ A clear relationship can be seen	☐ No clear relationship can be seen
☐ Repeats were done and averaged	☐ No repeats were done
☐ The measuring instruments worked well	☐ Something went wrong with the instruments, or with doing the experiment
☐ All the variables were controlled	☐ Some variables were hard to control

Write it like this

My method was **valid** / **partly valid** / **not valid**,
because _____.

Write about the **design and the data**. Working slowly, running out of time, or a single slip are not what this question is about.

My method was partly valid. My points lie close to a smooth curve and a clear relationship can be seen, and I repeated every reading three times and averaged them. The reading at 60° sits below the curve, which suggests the ray was not quite on the centre of the card for that measurement.

C5 Improvements and extensions

Improvements — two or more

An improvement makes the **data** better. Four questions to work through:

- (a) Could you **widen the range**? What happens at much bigger or much smaller values?
- (b) Could you take readings in **smaller steps** — going up in 5° rather than 10°?
- (c) Was anything **hard to measure accurately**? How would you fix it?
- (d) Is there a **more precise instrument** — one that would show a smaller change?

This is about the experiment, not about how fast you worked.

*A **narrower beam**. The light beam was wide enough that it was hard to say exactly where the middle of the ray crossed the scale. A narrower slit would give a sharper line and a reading I could trust to the nearest degree. **Smaller steps**. Going up in 5° instead of 10° would give eighteen points instead of nine, so the shape of the curve would be much clearer — especially near 80° where it flattens.*

Extension — one new investigation

An extension asks a **new research question** with a **different** independent variable. Two things to write:

1. What the new research question would be.
2. Briefly, how you would do it.

How does changing the colour of the light affect the angle of refraction? I would keep the angle of incidence fixed at 50° and put a different coloured filter in front of the ray box each time — red, green, then blue — reading the angle of refraction three times for each and averaging. Everything else would stay the same: the same block, the same card, the same power setting.

Before you hand it in

- ☐ B1 research question names **both** variables
- ☐ B2 hypothesis has a **because**
- ☐ B3 has a labelled diagram, values, resolutions, controls, repeats and safety
- ☐ C1 table has units **in the headings** and the averages worked out
- ☐ C1 graph has both axes labelled with units
- ☐ C2 conclusion uses **numbers from your own graph**
- ☐ C3 and C4 both say **because**
- ☐ C5 has **two** improvements and **one** extension